

# Problem Set 0 solutions

Due: Not required to hand in

CSE5825 BML

## 1 Probability

1. **Biting goats.** We use Bayes Theorem here. We know the probability of biting given that the goat is mean. We wish to know the probability that a goat is mean given that it bit someone.

$$P(\text{mean}|\text{bite}) = \frac{P(\text{bite}|\text{mean})P(\text{mean})}{P(\text{bite})} = \frac{P(\text{bite}|\text{mean})P(\text{mean})}{P(\text{bite}|\text{mean})P(\text{mean}) + P(\text{bite}|\text{nice})P(\text{nice})}$$

We have to make additional assumptions to compute this. We make assumptions on the prior probabilities of  $P(\text{mean})$  and  $P(\text{bite})$ , or the probabilities of a mean goat and biting goat before we receive any information about them. If mean and biting goats are common, e.g.,  $P(\text{mean}) = 0.5$  and  $P(\text{bite}) = 0.5$ , then we are confident about the status of the goat:  $P(\text{mean}|\text{bite}) = 0.8 \cdot 0.5/0.5 = 0.8$ . However it's less likely that a goat is mean, and even less likely to get bitten, e.g.  $P(\text{mean}) = 0.2$  and  $P(\text{bite}) = 0.1$ , and so  $P(\text{mean}|\text{bite}) = 0.8 \cdot 0.1/0.2 = 0.4$ , which makes us much less sure about the goat's attitude.

2. Proving properties of 4 random variables.

- (a) Prove that  $(X \perp\!\!\!\perp Y, W|Z) \Rightarrow (X \perp\!\!\!\perp Y|Z)$ . Note that if  $P(Z = z) = 0$  then the statement holds trivially and  $(X \perp\!\!\!\perp Y, W|Z)$  implies  $P(X, Y, W|Z) = P(X|Z)P(Y, W|Z)$ . Then,

$$\begin{aligned} P(X, Y|Z) &= \sum_w P(X, Y, w|Z) && \text{summing over } w \\ &= P(X|Z) \sum_w P(Y, w|Z) && \text{factor the joint, } P(X|Z) \text{ ind. of } w \\ &= P(X|Z)P(Y|Z) && \text{sum out } w \end{aligned}$$

- (b) Prove that  $(X \perp\!\!\!\perp Y, W|Z) \Rightarrow (X \perp\!\!\!\perp Y|W, Z)$ . We begin by considering the edge cases where variable assignments have probability 0. If  $P(W = w, Z = z) = 0$ , then  $(X \perp\!\!\!\perp Y|W, Z)$  holds trivially. It also holds if  $P(Z = z) = 0$  because  $P(Z = z) = 0$  implies  $P(W = w, Z = z) = 0$ . The more interesting case is when  $P(Z = z) \neq 0$  and  $P(W = w, Z = z) \neq 0$ .

$$\begin{aligned} P(X, Y|Z, W) &= P(X, Y, W|Z)/P(W|Z) && \text{cond. independence with } P(W|Z) \neq 0 \\ &= P(X|Z)P(Y, W|Z)/P(W|Z) && \text{given our assumption} \\ &= P(X|Z)P(Y|Z, W) && \text{factorizing } P(Y, W|Z) \text{ and cancelling} \end{aligned}$$

## 2 A simple model

1. What is  $MAP(A)$  (i.e.  $\arg \max_a(P(a))$ )?  $a^1$
2. What is  $MAP(B)$ ? (i.e.  $\arg \max_b(P(b))$ )?  
 $P(b^0) = P(b^0|a^0)P(a^0) + P(b^0|a^1)P(a^1) = 0.1 * 0.4 + 0.5 * 0.6 = 0.34$ .  
 $P(b^1) = P(b^1|a^0)P(a^0) + P(b^1|a^1)P(a^1) = 0.9 * 0.4 + 0.5 * 0.6 = 0.66$ .  
 So,  $MAP(B) = b^1$ .
3. What is  $MAP(A, B)$ ? (i.e.  $\arg \max_{a,b}(P(a, b))$ )?  
 $P(b^1|a^0)P(a^0) = 0.36$  and so  $MAP(A, B) = (a^0, b^1)$ . Notice how the MAP of the joint is distinct from the MAP of the individual marginals.

## 3 Conditional independencies

1. prove that  $(S \perp\!\!\!\perp D, G, L|I)$ . To prove this we show that  $P(S|I, D, G, L) = P(S|I)$ .

$$\begin{aligned}
 P(S|I, D, G, L) &= \frac{P(S, I, D, G, L)}{P(I, D, G, L)} \\
 &= \frac{P(I)P(D)P(G|I, D)P(L|G)P(S|I)}{P(I, D, G, L)} && \text{factorizing numerator} \\
 &= \frac{P(I)P(D)P(G|I, D)P(L|G)P(S|I)}{\sum_s P(I, D, G, L, s)} && \text{marginalizing s} \\
 &= \frac{P(I)P(D)P(G|I, D)P(L|G)P(S|I)}{P(I)P(D)P(G|I, D)P(L|G) \sum_s P(s|I)} && \text{factorizing denominator} \\
 &= \frac{P(I)P(D)P(G|I, D)P(L|G)P(S|I)}{P(I)P(D)P(G|I, D)P(L|G)} && \text{sum equal to 1} \\
 &= P(S|I)
 \end{aligned}$$

2. is  $(D \perp\!\!\!\perp I|G)$  true? No, this is a collider.
3. is  $(D \perp\!\!\!\perp I)$  true? Yes.
4. is  $(D \perp\!\!\!\perp I|L)$  true? No, this is a collider.

## 4 A million dollar app idea

This is a naive Bayes model.

1. How is  $C$  distributed? A categorical random variable.

2. How are the  $X_i$  for  $i = 1, \dots, n$  distributed? This depends on our likelihood model. For this problem, we could assume that the  $X_i$  are the words in the  $i^{th}$  text, and so are distributed as a Dirichlet-Multinomial.
3. Factorize the joint probability  $P(C, X_1, \dots, X_n)$ .  $P(C, X_1, \dots, X_n) = P(C) \prod_{i=1}^n P(X_i|C)$  where this follows from the naive Bayes independence assumptions.
4. What distribution does the posterior follow? With these assumptions, a Dirichlet distribution.
5. Is this a good model for your particular problem? All models are wrong, some are useful! This model could probably benefit from some additional data about the sender or an  $n$ -gram model.

## 5 CpG Islands

See the implementation of max product on HuskyCT. Sum-product is a simple modification of max-product.